

The Balanced Doubling-Map Case for Random Products of Weighted Shifts

arXiv automatic math run `fa_banach_001`

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Status

This packet gives a full solution candidate, likely valid pending expert review, for the balanced doubling-map case left open in Gillet [1]. The conclusion is that, on $X = \ell_p$, $1 \leq p < \infty$, and on $X = c_0$, the random products in the source example are universal/weakly mixing almost surely, but not mixing almost surely.

Source Question

The source paper is Valentin Gillet, *Linear dynamics of random products of weighted shifts*, arXiv:2511.19161. In Section 4a, after treating the cases $\mu(A_1) > \mu(A_2)$ and $\mu(A_1) < \mu(A_2)$, the paper leaves open the balanced case for the doubling map:

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It remains unknown of what happens in the case where $\mu(A_1) = \mu(A_2) = 1/2$, even if $\Omega = \mathbb{T}$, τ is the doubling map on Ω , $A_1 = [0, 1/2)$ and $A_2 = [1/2, 1)$. In the latter case, the random variables $\mathbb{1}_{A_1} \circ \tau^l$, $l \geq 0$, are independent and have the same distribution, namely $\nu := \frac{1}{2}(\delta_0 + \delta_1)$. Indeed, let $\varphi : \{0, 1\}^{\mathbb{Z}_+} \rightarrow \mathbb{T}$ be given by $\varphi((x_j)_{j \geq 0}) = \sum \frac{x_j}{2^{j+1}}$ and

We answer this exact balanced case.

Setup and Source Criterion

Let $\Omega = \mathbb{T}$ with normalized Lebesgue measure, let $\tau x = 2x \pmod{1}$, and set

$$A_1 = [0, 1/2), \quad A_2 = [1/2, 1).$$

Let $X = \ell_p$, $1 \leq p < \infty$, or $X = c_0$. Define weighted shifts by

$$w_l = 1 + \frac{1}{l+1}, \quad v_l = 1 - \frac{1}{l+1} \quad (l \geq 1),$$

and

$$T(\omega) = \begin{cases} B_w, & \omega \in A_1, \\ B_v, & \omega \in A_2. \end{cases}$$

Write

$$T_n(\omega) = T(\tau^{n-1}\omega) \cdots T(\tau\omega)T(\omega).$$

By the source paper's criterion for $X = \ell_p$ and $X = c_0$, the sequence $(T_n(\omega))$ is universal, equivalently weakly mixing, iff there is an increasing sequence n_i such that for every fixed $j \geq 0$,

$$\prod_{l=1}^{n_i+j} \left| \varepsilon_l(\tau^{n_i+j-l}\omega) \right| \longrightarrow \infty.$$

It is mixing iff the same product tends to ∞ along the full sequence n .

Proof Intuition

In the balanced doubling-map case the variables

$$\xi_l(\omega) = 2\mathbf{1}_{A_1}(\tau^l\omega) - 1$$

are independent Rademacher variables. The logarithm of the product controlling universality is

$$V_n(\omega) = d_n + \frac{1}{2} \sum_{k=1}^n b_k \xi_{n-k}(\omega), \quad b_k = \log \left(1 + \frac{2}{k} \right),$$

where d_n is bounded and converges. Since $b_k \sim 2/k$, the coefficients are square summable but not summable. Thus the associated two-sided stationary Rademacher series converges, is ergodic under the shift, and has unbounded support in both directions. Ergodicity forces arbitrarily large positive and negative values with positive frequency. A short tail estimate transfers this back to the one-sided finite sums V_n .

Consequently V_n has $\limsup V_n = +\infty$ and $\liminf V_n = -\infty$ almost surely. The positive limsup gives the subsequence required for universality. The negative liminf prevents mixing.

The Result

Theorem 1. *In the balanced doubling-map setup above, for $X = \ell_p$, $1 \leq p < \infty$, and for $X = c_0$, the random product sequence $(T_n(\omega))_{n \geq 1}$ is universal and weakly mixing for Lebesgue-almost every $\omega \in \mathbb{T}$. It is not mixing for Lebesgue-almost every $\omega \in \mathbb{T}$.*

Proof. Let

$$X_l(\omega) = \mathbf{1}_{A_1}(\tau^l\omega), \quad \xi_l(\omega) = 2X_l(\omega) - 1.$$

Under the binary coding of the doubling map, the ξ_l 's are independent Rademacher variables. Put

$$b_k = \log \left(1 + \frac{2}{k} \right), \quad V_n(\omega) = \log \prod_{l=1}^n \left| \varepsilon_l(\tau^{n-l}\omega) \right|.$$

The computation in the source paper gives

$$V_n = -\log(n+1) + \sum_{k=1}^n b_k X_{n-k} = d_n + \frac{1}{2} \sum_{k=1}^n b_k \xi_{n-k},$$

where

$$d_n = -\log(n+1) + \frac{1}{2} \sum_{k=1}^n b_k = -\log(n+1) + \frac{1}{2} \log \frac{(n+1)(n+2)}{2} \longrightarrow -\frac{1}{2} \log 2.$$

We first prove

$$\limsup_{n \rightarrow \infty} V_n = +\infty, \quad \liminf_{n \rightarrow \infty} V_n = -\infty \quad \text{almost surely.}$$

Work for the moment on the two-sided Bernoulli shift $(\{-1, 1\}^{\mathbb{Z}}, \nu, \sigma)$, and denote the coordinates again by ξ_j . Since $\sum b_k^2 < \infty$, the Rademacher series

$$Y_n = d + \frac{1}{2} \sum_{k=1}^{\infty} b_k \xi_{n-k}, \quad d = -\frac{1}{2} \log 2,$$

converges almost surely and in L^2 . The process $Y_n = Y_0 \circ \sigma^n$ is stationary and ergodic.

The law of Y_0 has unbounded support above and below. Indeed, $\sum b_k = \infty$. Given $M > 0$, choose K so large that

$$d + \frac{1}{2} \sum_{k=1}^K b_k > M + L$$

for some L with

$$\nu \left\{ \frac{1}{2} \sum_{k>K} b_k \xi_{-k} > -L \right\} > 0.$$

On the positive-probability event where $\xi_{-1} = \dots = \xi_{-K} = 1$ and the tail is larger than $-L$, one has $Y_0 > M$. The lower tail is identical, using $\xi_{-1} = \dots = \xi_{-K} = -1$. Hence

$$\nu(Y_0 > M) > 0, \quad \nu(Y_0 < -M) > 0 \quad (M > 0).$$

By Birkhoff's ergodic theorem, for almost every two-sided sequence the sets $\{n : Y_n > M\}$ and $\{n : Y_n < -M\}$ have positive lower frequency.

It remains to transfer this statement from Y_n to V_n . On the two-sided extension,

$$Y_n - V_n = (d - d_n) + \frac{1}{2} \sum_{r=1}^{\infty} b_{n+r} \xi_{-r}.$$

Therefore

$$\mathbb{E}|Y_n - V_n|^2 \leq C|d - d_n|^2 + C \sum_{r=1}^{\infty} b_{n+r}^2 \leq \frac{C'}{n}.$$

Consequently

$$\mathbb{E} \left[\frac{1}{N} \sum_{n=1}^N |Y_n - V_n|^2 \right] \leq C'' \frac{\log N}{N} \longrightarrow 0.$$

Applying this estimate along dyadic N 's and using Borel–Cantelli gives

$$\frac{1}{N} \sum_{n=1}^N |Y_n - V_n|^2 \longrightarrow 0 \quad \text{almost surely.}$$

Thus, for every fixed $a > 0$, the set of n such that $|Y_n - V_n| > a$ has zero asymptotic density. Combining this zero-density exceptional set with the positive-frequency visits of Y_n above $M + 1$ and below $-M - 1$, we obtain

$$\limsup_{n \rightarrow \infty} V_n = +\infty, \quad \liminf_{n \rightarrow \infty} V_n = -\infty$$

for almost every two-sided sequence. Since this assertion depends only on the nonnegative coordinates, it holds for almost every point of the original one-sided doubling system.

Choose an increasing sequence n_i such that $V_{n_i} \rightarrow +\infty$. For each fixed $j \geq 0$, the source criterion needs $V_{n_i+j} \rightarrow +\infty$. This follows because the drop from V_n to V_{n+j} is bounded below by a constant depending only on j . Indeed, using the formula

$$V_n = -\log(n+1) + \sum_{k=1}^n b_k X_{n-k}$$

and the monotonicity of b_k ,

$$\begin{aligned} V_{n+j} - V_n &\geq -\log \frac{n+j+1}{n+1} - \sum_{k=1}^n (b_k - b_{k+j}) X_{n-k} \\ &\geq -\log(j+1) - \sum_{k=1}^{\infty} (b_k - b_{k+j}) \\ &= -\log(j+1) - \sum_{k=1}^j b_k. \end{aligned}$$

Therefore $V_{n_i+j} \rightarrow +\infty$ for every fixed $j \geq 0$. By the source criterion, $(T_n(\omega))$ is universal and weakly mixing almost surely.

On the other hand, $\liminf V_n = -\infty$ implies in particular that V_n does not tend to $+\infty$ along the full sequence. The source mixing criterion therefore shows that $(T_n(\omega))$ is not mixing almost surely. $\square$

Verification and Novelty Check

The proof uses the source paper's exact universality/mixing criteria for ℓ_p and c_0 , the binary independence of the doubling map already identified in the source, and standard facts about square-summable Rademacher series and ergodic stationary processes.

Local run indexes were searched for arXiv:2511.19161, random products of weighted shifts, balanced doubling, Nörlund sums, and related terms; no duplicate packet was found. External web searches on July 3, 2026 for the arXiv id, the exact title, the exact balanced-case phrase, and combinations of “Norlund”, “doubling map”, “weighted shifts”, and “universal” did not surface a separate answer. The arXiv page for [1] records version 2 from December 15, 2025 and its abstract describes criteria plus examples, not the balanced-case solution proved here.

Human Review Recommendation

This should be reviewed as a full answer to the specific balanced doubling-map case for $X = \ell_p$ and $X = c_0$. The main proof point to audit is the transfer from the two-sided stationary comparison

process to the finite one-sided sums V_n , especially the density-zero estimate for $|Y_n - V_n| > a$. The conclusion does not claim to settle balanced cases for arbitrary ergodic systems or arbitrary partitions.

References

- [1] V. Gillet, *Linear dynamics of random products of weighted shifts*, arXiv:2511.19161, 2025.